

This lab consists of FIVE Practical

Practical-1:

- Demonstrate sampling using R

Sampling in R

The **sample** function is used to generate a random sample from a given population. It can be used to sample with or without replacement by using the **replace** argument

```
> sample(1:20, 10, replace=TRUE)
```

```
[1] 5 16 4 7 7 1 8 12 14 18
```

```
> sample(1:20, 10)
```

```
[1] 7 14 18 1 4 6 17 5 20 3
```

Sample Function in R with set.seed():

If we run the sample function again and again, it gives different set of samples each time. What if we require the same sample each time?

For that we use [set.seed\(\)](#) function. Set.seed() function always returns the same pseudo-random sequence.

```
3 set.seed(10)
```

```
4 sample(1:20, 10, replace=TRUE)
```

Whenever we run the sample function along with the [set.seed\(\) function](#) it outputs same set of samples. so the output always will be the same for set.seed(10)

```
[1] 11 7 9 14 2 5 6 6 13 9
```

Sample Function in R with dataset:

```
> mtcars
```

	mpg	cyl	dis	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	0	3	3
Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3
Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	0	3	3
Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6
Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	1	5	8
Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2

```

> set.seed(123)
> index<-sample(1:nrow(mtcars), 5)
> index
[1] 10 25 13 26 27
> mtcars[index,]
      mpg cyl  disp  hp drat   wt  qsec vs am gear carb
Merc 280    19.2   6 167.6 123 3.92 3.440 18.30  1  0   4    4
Pontiac Firebird 19.2   8 400.0 175 3.08 3.845 17.05  0  0   3    2
Merc 450SL    17.3   8 275.8 180 3.07 3.730 17.60  0  0   3    3
Fiat X1-9     27.3   4  79.0  66 4.08 1.935 18.90  1  1   4    1
Porsche 914-2  26.0   4 120.3  91 4.43 2.140 16.70  0  1   5    2

```

- sample function returns the 5 row indexes to the index object
- Then that 5 indexes are passed as input to the mtcars to fetch that 5 rows

So the output will be

```
[1] 10 25 13 26 27
```

and the sample dataframe will be

```

      mpg cyl  disp  hp drat   wt  qsec vs am gear carb
Merc 280    19.2   6 167.6 123 3.92 3.440 18.30  1  0   4    4
Pontiac Firebird 19.2   8 400.0 175 3.08 3.845 17.05  0  0   3    2
Merc 450SL    17.3   8 275.8 180 3.07 3.730 17.60  0  0   3    3
Fiat X1-9     27.3   4  79.0  66 4.08 1.935 18.90  1  1   4    1
Porsche 914-2  26.0   4 120.3  91 4.43 2.140 16.70  0  1   5    2

```

As we have set seeds it will give the same set of 5 rows as a sample.

Practical-2:

- Demonstrate sampling distribution using R

Sampling Distribution in R

1. Generating random samples from a normal distribution

Even though we would like to think of our samples as random, it is in fact almost impossible to generate random numbers on a computer. So, we will admit that we are really drawing a pseudo-random sample. In order to be able to reproduce the results we will set the seed for our pseudo-random number generator to the value of 124 using the **set.seed** function.

```
set.seed(124)
```

It is often very useful to be able to generate a sample from a specific distribution. To generate a sample of size 100 from a standard normal distribution (with mean 0 and standard deviation 1) we use the **rnorm** function. We only have to supply the **n** (sample size) argument since mean 0 and standard deviation 1 are the default values for the **mean** and **stdev** arguments.

```
norm <- rnorm(100)
```

Now let's look at the first 10 observations. We use square brackets to surround the first and last element number. In the output, the number of the first element listed on the line is given in the square brackets. For example, the [9] indicates that the first number given (0.19709386) is the ninth element.

```

norm[1:10]
## [1] -1.3851  0.0383 -0.7630  0.2123  1.4255  0.7445  0.7002 -0.2294
## [9]  0.1971  1.2072

```

```
mean(norm)
```

```
## [1] 0.00962
```

```
sd(norm)
```

```
## [1] 0.884
```

If we want to obtain a sample of values drawn from a normal distribution with a different value for the mean and standard deviation then we just have to use the **mean** and **sd** arguments.

Let's draw a sample of size 100 from a normal distribution with mean 2 and standard deviation 5.

```
>set.seed(124)
```

```
>norm <- rnorm(100, 2, 5)
```

```
>norm
```

```
>norm[1:10]
```

```
>mean(norm) //check mean
```

```
>sd(norm)
```

```
# random sample of size 10 from sequence [1, 5] with unequal probabilities# of being chosen
```

```
>set.seed(124)
```

```
>sample(5, 10, prob = c(0.3, 0.4, 0.1, 0.1, 0.1), replace = T)
```

```
[1] 2 1 1 2 2 2 1 1 4 2
```

Exercise :

In a sample of 16 observations from a normal distribution with mean 150 and standard deviation 16 ,

1. What is $P(\bar{x} \leq 160)$? $P(\bar{x} \leq 162)$ - lower.tail=T

2. What is $P(\bar{x} > 142)$?

```
> Sample1=rnorm(16, 150, 16)
```

```
> Sample1
```

```
[1] 169.0674 162.2751 142.1420 168.6629 171.3160 146.7353 136.6156 138.1796
```

```
[9] 132.1003 166.2931 147.4985 157.9000 137.7103 136.6380 164.3930 160.5710
```

```
> mean(Sample1)
```

```
[1] 152.3811
```

```
> sd(Sample1)
```

```
[1] 13.95347
```

```
> sigma_xbar=16/4
```

1. What is $P(\bar{x} \leq 160)$?

```
> z=(160-150)/4
```

```
> z
```

```
[1] 2.5
```

```
> pnorm(z)
```

```
[1] 0.9937903
```

```
> pnorm(z, lower.tail=TRUE)
```

```
[1] 0.9937903
```

2. What is $P(\bar{x} > 142)$?

```
zscore=(142-150)/sigmaxbar
```

```
> zscore
```

```
> -2
```

```
> pnorm(zscore)
```

```
[1] 0.02275013
```

```
> pnorm(zscore, lower.tail=FALSE)
```

```
[1] 0.9772499
```

Pnorm(): pnorm is the R function that calculates the c. d. f

```
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE, log.p = FALSE)
```

Q- vector of quantiles.

Mean-vector of means.

Sd-vector of standard deviations.

log, log.p-logical; if TRUE, probabilities p are given as $\log(p)$.

lower.tail -logical; if TRUE (default), probabilities are $P[X \leq x]$ otherwise, $P[X > x]$.

$F(x) = P(X \leq x)$ where X is normal.

Both of the R commands given below do exactly the same thing.

```
pnorm(27.4, mean=50, sd=20)
```

```
pnorm(27.4, 50, 20)
```

They look up $P(X < 27.4)$ when X is normal with mean 50 and standard deviation 20.

Example: Suppose widget weights produced at Acme . Widget Works have weights that are normally distributed with mean 17.46 grams and variance 375.67 grams. What is the probability that a randomly chosen widget weighs more than 19 grams?

Question Rephrased: What is $P(X > 19)$ when X has the $N(17.46, 375.67)$ distribution?

```
> 1 - pnorm(19, mean=17.46, sd=sqrt(375.67))
```

Practical-3

- **Exercises: based on sampling and sampling distribution**

Find commands and output for following exercises:

1. A bottling company uses a filling machine to fill plastic bottles with a popular cola. The bottles are supposed to contain 300 ml. In fact, the contents vary according to a normal distribution with a mean (μ) = 298 ml and standard deviation (σ) = 3 ml. What is the probability that the average contents of 6 randomly selected bottles is < 295 ?

2. Consider Mean = 8000 and standard deviation is = 3200 , n=64 , what is $P(\bar{x} > 9000)$ and $P(\bar{x} \leq 8500)$

3. In a sample of observations from a normal distribution with mean 18 and standard deviation 4.8 ,

- i. What is $P(\bar{x} \text{ less than or equal to } 16)$
- ii. What is $P(\bar{x} > 16)$
- iii. What is $P(\bar{x} \leq 20)$

Practical-4:

- **Demonstrate Hypothesis Testing using R**

Theory: A hypothesis test is a statistical test that is used to determine whether there is enough evidence in a sample of data to infer that a certain condition is true for the entire population. A hypothesis test examines two opposing hypotheses about a population: the null hypothesis and the alternative hypothesis.

Hypothesis testing begins with an assumption called ‘Hypothesis’ that we make about a population parameter.

Steps in hypothesis testing

1. We collect sample data , calculate sample statistics and then use this information to decide how likely it is that our hypothesized population parameter(say population mean) is correct .
2. To test the validity of our assumption , we gather sample data and determine the difference between the hypothesized value and actual value of sample mean
3. Then we judge whether the difference is significant
4. If difference is smaller, the greater the likelihood that our hypothesized value is correct

With one sample, Hypothesis tests can be of different types :

1. *Two-Tailed Test of Population Mean with Known Variance*
2. *left-Tailed Test of Population Mean with Known Variance*
3. *right-Tailed Test of Population Mean with Known Variance*
4. *Two-Tailed Test of Population Mean with Unknown Variance*
5. *left-Tailed Test of Population Mean with Unknown Variance*
6. *right-Tailed Test of Population Mean with Unknown Variance*

1. Two-Tailed Test of Population Mean with Known Variance in R:

The null hypothesis of the two-tailed test of the population mean can be expressed as follows:

$$\mu = \mu_0$$

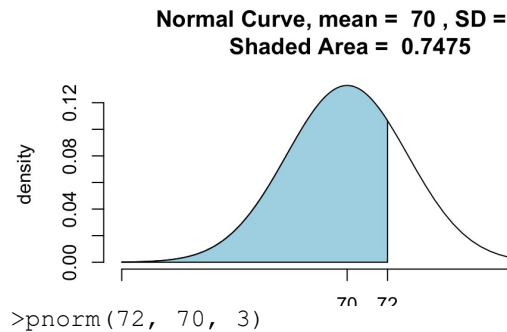
where μ_0 is a hypothesized value of the true population mean μ .

Let us define the test statistic z in terms of the sample mean, the sample size and the population standard deviation σ :

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Then the null hypothesis of the two-tailed test is to be rejected if $z \leq -z_{\alpha/2}$ or $z \geq z_{\alpha/2}$, where $z_{\alpha/2}$ is the 100(1 - $\alpha/2$) percentile of the standard normal distribution.

qnorm() : The function **pnorm()** in regular R, compute probabilities from known bounding values. For example, suppose that X is a normally distributed random variable with mean 70 and standard deviation 3, and that you want to know: $P(X < 72)$. Then you *know* the boundary value 72, but you don't know the probability: the area under the normal density curve before 72.



The function **qnorm()**, which comes standard with R, aims to find **given an area, find the boundary value that determines this area.**

For example, suppose you want to find that 85th percentile of a normal distribution whose mean is 70 and whose standard deviation is 3. Then you ask for:

```
qnorm(0.85, mean=70, sd=3)
## [1] 73.1093
```

The value 73.1093 is indeed the 85th percentile, in the sense that 85% of the values in a population that is normally distributed with mean 70 and standard deviation 3 will lie below 73.1093. In other words, if you were to pick a random member X from such a population, then

$$P(X < 73.1093) = 0.85$$

Problem

Suppose the mean weight of King Penguins found in an Antarctic colony last year was 15.4 kg. **In a sample of 35 penguins same time this year in the same colony, the mean penguin weight is 14.6 kg.** Assume the population standard deviation is 2.5 kg. At .05 significance level, can we reject the null hypothesis that the mean penguin weight does not differ from last year?

Solution

The null hypothesis is that $\mu = 15.4$. We begin with computing the test statistic.

```
> xbar = 14.6      # sample mean
> mu0 = 15.4      # hypothesized value
> sigma = 2.5     # population standard deviation
> n = 35          # sample size
> z = (xbar - mu0) / (sigma / sqrt(n))
> z              # test statistic
[1] -1.8931
```

We then compute the critical values at .05 significance level.

```
> alpha = .05
> z.half.alpha = qnorm(1 - alpha/2)
> c(-z.half.alpha, z.half.alpha)
[1] -1.9600 1.9600
```

Answer

The test statistic -1.8931 lies between the critical values -1.9600 and 1.9600. Hence, at .05 significance level, we do not reject the null hypothesis that the mean penguin weight does not differ from last year.

Alternative Solution

Instead of using the critical value, we apply the pnorm function to compute the two-tailed p-value of the test statistic. It doubles the lower tail p-value as the sample mean is less than the hypothesized value. Since it turns out to be greater than the .05 significance level, we do not reject the null hypothesis that $\mu = 15.4$.

```
> pval = 2 * pnorm(z) # lower tail
> pval                # two-tailed p-value
[1] 0.058339
```

2. Lower Tail Test of Population Mean with Known Variance in R

The null hypothesis of the lower tail test of the population mean can be expressed as follows:

$$\mu \geq \mu_0$$

where μ_0 is a hypothesized lower bound of the true population mean μ .

Let us define the test statistic z in terms of the sample mean, the sample size and the population standard deviation σ :

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Then the null hypothesis of the lower tail test is to be rejected if $z \leq -z_\alpha$, where z_α is the $100(1 - \alpha)$ percentile of the standard normal distribution.

Problem

Suppose the manufacturer claims that the mean lifetime of a light bulb is more than 10,000 hours. In a sample of 30 light bulbs, it was found that they only last 9,900 hours on average. Assume the population standard deviation is 120 hours. At .05 significance level, can we reject the claim by the manufacturer?

Solution

The null hypothesis is that $\mu \geq 10000$. We begin with computing the test statistic.

```
> xbar = 9900      # sample mean
> mu0 = 10000      # hypothesized value
> sigma = 120      # population standard deviation
> n = 30           # sample size
> z = (xbar-mu0)/(sigma/sqrt(n))
> z               # test statistic
[1] -4.5644
```

We then compute the critical value at .05 significance level.

```
> alpha = .05
> z.alpha = qnorm(1-alpha)
```

```
> -z.alpha          # critical value
[1] -1.6449
```

Answer

The test statistic -4.5644 is less than the critical value of -1.6449. Hence, at .05 significance level, we reject the claim that mean lifetime of a light bulb is above 10,000 hours.

Alternative Solution

Instead of using the critical value, we apply the pnorm function to compute the lower tail p-value of the test statistic. As it turns out to be less than the .05 significance level, we reject the null hypothesis that $\mu \geq 10000$.

```
> pval = pnorm(z)
> pval          # lower tail p-value
[1] 2.5052e-06
```

3. Two-Tailed Test of Population Mean with Unknown Variance

The null hypothesis of the two-tailed test of the population mean can be expressed as follows:

$$\mu = \mu_0$$

where μ_0 is a hypothesized value of the true population mean μ .

Let us define the test statistic t in terms of the sample mean, the sample size and the sample standard deviation s :

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Then the null hypothesis of the two-tailed test is to be rejected if $t \leq -t_{\alpha/2}$ or $t \geq t_{\alpha/2}$, where $t_{\alpha/2}$ is the $100(1 - \alpha)$ percentile of the Student t distribution with $n - 1$ degrees of freedom.

Problem

Suppose the mean weight of King Penguins found in an Antarctic colony last year was 15.4 kg. In a sample of 35 penguins same time this year in the same colony, the mean penguin weight is 14.6 kg. Assume the sample standard deviation is 2.5 kg. At .05 significance level, can we reject the null hypothesis that the mean penguin weight does not differ from last year?

Solution

The null hypothesis is that $\mu = 15.4$. We begin with computing the test statistic.

```
> xbar = 14.6          # sample mean
> mu0 = 15.4           # hypothesized value
> s = 2.5              # sample standard deviation
> n = 35               # sample size
> t = (xbar-mu0)/(s/sqrt(n))
> t                    # test statistic
[1] -1.8931
```

We then compute the critical values at .05 significance level.

```
> alpha = .05
> t.half.alpha = qt(1-alpha/2, df=n-1)
```

```
> c(-t.half.alpha, t.half.alpha)
[1] -2.0322 2.0322
```

Answer

The test statistic -1.8931 lies between the critical values -2.0322, and 2.0322. Hence, at .05 significance level, we do not reject the null hypothesis that the mean penguin weight does not differ from last year.

Alternative Solution

Instead of using the critical value, we apply the pt function to compute the two-tailed p-value of the test statistic. It doubles the lower tail p-value as the sample mean is less than the hypothesized value. Since it turns out to be greater than the .05 significance level, we do not reject the null hypothesis that $\mu = 15.4$.

```
> pval = 2 * pt(t, df=n-1) # lower tail
> pval                      # two-tailed p-value
[1] 0.066876
```

4. Upper Tail Test of Population Mean with Unknown Variance

The null hypothesis of the upper tail test of the population mean can be expressed as follows:

$$\mu \leq \mu_0$$

where μ_0 is a hypothesized upper bound of the true population mean μ .

Let us define the test statistic t in terms of the sample mean, the sample size and the sample standard deviation s :

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Then the null hypothesis of the upper tail test is to be rejected if $t \geq t_\alpha$, where t_α is the 100(1 - α) percentile of the Student t distribution with $n - 1$ degrees of freedom.

Problem

Suppose the food label on a cookie bag states that there is at most 2 grams of saturated fat in a single cookie. In a sample of 35 cookies, it is found that the mean amount of saturated fat per cookie is 2.1 grams. Assume that the sample standard deviation is 0.3 gram. At .05 significance level, can we reject the claim on food label?

Solution

The null hypothesis is that $\mu \leq 2$. We begin with computing the test statistic.

```
> xbar = 2.1          # sample mean
> mu0 = 2             # hypothesized value
> s = 0.3             # sample standard deviation
> n = 35              # sample size
> t = (xbar-mu0)/(s/sqrt(n))
> t                  # test statistic
[1] 1.9720
```

We then compute the critical value at .05 significance level.

```
> alpha = .05
> t.alpha = qt(1-alpha, df=n-1)
> t.alpha          # critical value
[1] 1.6991
```

Answer

The test statistic 1.9720 is greater than the critical value of 1.6991. Hence, at .05 significance level, we can reject the claim that there is at most 2 grams of saturated fat in a cookie.

Alternative Solution

Instead of using the critical value, we apply the pt function to compute the upper tail p-value of the test statistic. As it turns out to be less than the .05 significance level, we reject the null hypothesis that $\mu \leq 2$.

```
> pval = pt(t, df=n-1, lower.tail=FALSE)
> pval          # upper tail p-value
[1] 0.028393
```

Exercises for practical

Exercise1: HITAVADA press hypothesizes that average life of its largest web press is 14500 hours. They know standard deviation of press life is 2100 hours. From a sample of 25 presses, the company finds a sample mean of 13000 hours. At a 0.01 significance level, should the company conclude that average life of presses is less than hypothesized 14500 hours?

Exercise 2: Given a sample mean of 83, a sample standard deviation of 12.5, and a sample size of 22, test the hypothesis that the value of the population mean is 70 against the alternative that it is more than 70. Use the 0.025 significance level.

Exercise 3: Given a sample mean of 94.3, a sample standard deviation of 8.4, and a sample size of 6, test the hypothesis that the value of the population mean is 100 against the alternative that it is less than 100. Use the 0.05 significance level.

Practical-5:

- **Demonstrate one way ANOVA Testing using R**

Analysis of Variance (ANOVA) is a statistical technique, commonly used to study differences between two or more group means. ANOVA test is centered on the different sources of variation in a typical variable.

ANOVA in R primarily provides evidence of the existence of the mean equality between the groups. This statistical method is an extension of the t-test. It is used in a situation where the factor variable has more than one group.

ANOVA takes into account three types of variations - the **total sum of squares** (SST), the **sum of squares within groups** (SSW), and the **sum of squares between groups** (SSB).

- **H0** - All sample (group or factor) means are equal or they don't differ significantly.
- **H1** - At least one of the sample means is different from the rest.

```
df <-
data.frame(
  r1 = c(5, 8, 7, 3, 2),
  r2 = c(12, 8, 10, 10, 15),
  r3 = c(7, 8, 5, 4, 6)
)
>df_stacked <- stack(df[, 1:3])
>df_stacked
>anova <- aov(values ~ ind, data =
df_stacked)
>summary(anova)
```

The ANOVA in R function uses a P-value instead of comparing F-value to the critical value directly. It's just another way to interpret the results - commonly, if a P-value is below 0.05, we can say we're rejecting the null hypothesis in favor of the alternative one at a 95% confidence interval.

Question: implement following using R

A research company has designed three different systems to clean up oil spills. The following table contains the results, measured by how much surface area (in square meters) is cleared in 1 hour. The data were found by testing each method in several trials. Are the three systems equally effective. Use 0.05 level of significance

System A	55	60	63	56	59	55
System B	57	53	64	49	62	
System C	66	52	61	57		